



**AI Chat Bot**  
**Artificial Intelligence(AI101B)**  
**MCA 1<sup>st</sup> year (Sec-A)**  
**Session 2024 – 25**

**Submitted By:**

**Ashish Yadav      202410116100041**

**Aayush Sharma    202410116100005**

**Supervised by**  
**Mr. Apoorv Jain**  
**Assistant professor**

# Introduction

The **Ai Chat Bot** using a **Sequence-to-Sequence (Seq2Seq)** model, NLP and Transformers with **LSTM** units, which is commonly used for tasks like machine translation, text summarization, and speech recognition. The goal of the project is to answer the user about their questions.

Seq2Seq Translation Cycle



# Objective

- **Develop a functional AI chatbot system** capable of engaging in basic conversational interactions.
- **Implement the core principles of Natural Language Processing (NLP)** to enable the chatbot to understand user input and generate coherent responses.
- **Leverage the Transformer architecture** as the underlying neural network model for robust sequence-to-sequence (Seq2Seq) text generation.
- **Demonstrate the application of Seq2Seq models** in a conversational AI context for mapping user queries to appropriate replies.

# Chatbot Systems

A chatbot is a software application that simulates human conversation through text or voice interactions.

- **Types of Chatbots:**

- **Rule-Based:** Respond using predefined patterns or decision trees.
- **AI-Powered:** Use NLP and machine learning to understand and generate human-like responses.

- **Core Components:**

- **Natural Language Understanding (NLU)** – Interprets user input
- **Dialogue Manager** – Maintains context and decides the response
- **Natural Language Generation (NLG)** – Forms the reply in human language

- **Applications:**

- Customer support (e.g., banking, e-commerce)
- Virtual assistants (e.g., Siri, Google Assistant)
- Education, healthcare, entertainment

- **Technologies Used:**

- NLP frameworks (spaCy, NLTK)
- Deep learning models (Transformers, RNNs)
- APIs (Dialogflow, Rasa, Microsoft Bot Framework)

 Chatbots are transforming the way we interact with digital systems — making communication faster, smarter, and more human-like.

# Transformers

- **What are Transformers?**

A deep learning architecture introduced in 2017 (by Vaswani et al.) designed to process sequential data like language — without relying on recurrence (RNNs).

- **Key Concept:**

Uses **self-attention** to understand the relationships between all words in a sentence simultaneously.

- **Why Transformers for Chatbots?**

- Handle **long-range dependencies** in text
- Enable **context-aware** and **coherent** responses
- Power models like **GPT**, **BERT**, and **T5**

- **Core Components:**

- **Encoder–Decoder** structure (in models like T5)
- **Self-Attention Mechanism**
- **Positional Encoding**

- **Applications in Chatbots:**

- Natural conversation flow
- Context retention across turns

# Natural Language Processing

- **Definition:** NLP is a branch of artificial intelligence that enables machines to understand, interpret, and generate human language.
- **Key Tasks:**
  - Text classification (e.g., spam detection)
  - Sentiment analysis
  - Machine translation
  - Named entity recognition (NER)
  - Question answering
- **Applications:**
  - Virtual assistants (e.g., Siri, Alexa)
  - Goal: Bridge the gap between human communication (natural language) and computer understanding.
  - Chatbots
  - Search engines
  - Language translation tools (e.g., Google Translate)

# Understanding Seq2Seq Data Generation

- **Seq2Seq Model Structure**

The Seq2Seq model integrates both encoder and decoder mechanisms to facilitate data generation.

- A **Sequence-to-Sequence (Seq2Seq)** model takes a sequence (e.g., English sentence) as input and outputs another sequence (e.g., French sentence).

- It is commonly used for **machine translation, chatbots, and text summarization**.

# Creating a Seq2Seq Translation Model

## Core Architecture

- **Encoder**  
Converts input sentence into a fixed-length context vector using LSTM.
- **Decoder**  
Takes the context vector and generates the target sentence in French.

## Key Components

- **Embedding Layers:** Learn word representations.
- **LSTM Units:** Capture sequential dependencies.
- **Dense + Softmax:** Predict next word in the sequence.

## Data Flow

1. English sentence → Encoder → Context vector
2. Context vector + French input → Decoder → French output

## Goal

- Build a neural model that can learn sentence structure and meaning to generate accurate translations.



# Understanding Text Tokenization Techniques

## What is Tokenization?

- The process of breaking text into smaller units like **words** or **subwords**.
- Essential for converting human language into a form that models can process.

## Techniques Used

- **Word Tokenization:** Splits sentences into individual words.

*Example:* "I am happy" → ['I', 'am', 'happy']

- **Integer Encoding:** Maps each token to a unique number.

*Example:* 'happy' → 7

- **Padding:** Ensures all sequences have the same length for batch processing.

## Why It Matters?

- Enables consistent input size.
- Helps the model understand sentence structure.
- Crucial for learning context and word relationships.

Thank You!